AL-BATTĀNĪ

The Sabian Zīj

al-Zīj al-Ṣābi'

the canons, translated from the Arabic

in Nallino's edition, Part III (Rome, 1899)

This is an English translation of the canons of al-Battānī's *Zīj*, made from the Arabic text as Nallino printed it. Every sentence of the Arabic is translated, in the order of the text; the numbers are given in figures. Words added for the sense stand in square brackets. The notes identify quotations, record where the translation follows a correction of Nallino's, and explain terms. The pages of the Arabic text are given in the margin. Nallino's Latin translation (Part I, Milan, 1903) was used as a check on the meaning throughout; its wording is not followed.

The translation is in progress. This version holds the list of chapters and chapters 1, 2, 3, 4, 5, 6, 7, 8, 9. It is AI-integrated work (Claude), 2026; no person has proofread it. It is dedicated to the public domain (CC0 1.0).

Ar. p. 1

In the name of God, the Merciful, the Compassionate. God bless the Prophet Muḥammad, His noble Messenger, and his family and his companions, and grant them peace.

This is the whole of what Muḥammad ibn Jābir ibn Sinān al-Ḥarrānī, known as al-Battānī — may God pardon him — set down on the computation of the stars and on the places of their courses as they have been tested. The book contains fifty-seven chapters in all, and this is an account of them.[1]

1. The opening of the book.
2. On dividing the circle of the sphere, and on multiplying degrees by one another, taking their roots, and dividing them by one another.
3. On finding the lengths of the chords of the parts of the circle, on entering in tables the halves of the chords of double the arcs, and on the procedures with them that follow from this.[2]
4. On the amount of the inclination of the ecliptic to the equator; on the division of this inclination, its directions, and its stages as it rises and descends. It is the declination of the sun from the right sphere.[3]
5. On finding the amounts of the equator that rise together with given arcs of the ecliptic [in places] under the equator, which is called the line of equality. By these same amounts, too, the signs pass over the meridian circle in every place on the earth; this is why they are called the ascensions of the signs in the right sphere.[4]
6. On the properties of each of the lines parallel to the equator that lie away from it towards the north, and an account of the places of the inhabited earth whose longitude and latitude are known from the book *The Form of the Earth*. Ar. p. 2
7. On finding the amplitudes of the winter and summer risings and settings on the horizon circles of the localities. These are the arcs of the horizon between the equator and the places of the ecliptic, and they are called the azimuths of the risings and settings on the horizon circle.
8. On finding the altitude of the north pole from the excess of the longest day, when this is given.
9. On finding the excess of the longest day from the given altitude of the pole.
10. On finding the altitude and the shadow, each from the other, when the shadow is horizontal, and on finding them when the shadow is vertical.
11. On finding the azimuth of the altitude and of the shadow on the horizon circle, in every locality, at every time of the day, for all degrees of the ecliptic. This is the part of the horizon circle, counted from the limit of rising and setting, that is cut off by the arc passing through the zenith and the sun.
12. On finding the meridian line in every locality, which is the direction of the south, and the direction of the equinoctial rising that becomes known with it; and on finding it in various ways.

[1] The heading gives the author's name as «Muḥammad ibn Jābir ibn Sinān»; the title page that Nallino set above it has «ibn Sinān ibn Jābir». «Tested» (*al-mumtaḥan*) is the word al-Battānī uses for what has been verified by observation.

[2] The halves of the chords of double the arcs are the sines; chapter 3 introduces them under this name.

[3] The print has «ونجزية»; Nallino's corrigenda read «وتجزية», «the division», which is translated.

[4] Nallino suspects that the words «in the localities that are» have fallen out before «under the equator»; the bracket supplies their sense. «The line of equality» (*khaṭṭ al-istiwāʾ*) is the terrestrial equator.

13. On finding the amount of the equator that rises together with given degrees of the ecliptic from the horizon in every place on the earth, which is called the ascensions of the signs in each locality; and what follows from this: finding the ascensions of any portion you wish in these ascensions and in the ascensions of the right sphere, finding the degrees of the ecliptic from these ascensions, the amount of the arc of the day and of the night, their equal hours, the times of the seasonal hours of the day and of the night, and converting the one kind of hours into the other.[5]

14. On finding the latitudes of the localities, which is the altitude of the north pole above their horizon, by observation.

15. On finding the altitude of the sun at noon on any day.

16. On finding how many hours of the day have passed and what is rising, by measuring the sun and knowing the altitude and the vertical shadow.

17. On finding the altitude from the hours of the day that have passed.

18. On finding the distances of the fixed or the wandering stars from the equator when they lie away from the belt of the signs in latitude, and the degrees of the ecliptic that culminate with them, from their places in longitude and latitude with respect to the ecliptic. Ar. p. 3

19. On finding the half of the arc of the day of a star, which is half of its stay above the earth, and also below it, and the times of its hours above the earth and below it.[6]

20. On finding the degree of the ecliptic with which a star rises and the degree with which it sets.

21. On finding how many hours of the night have passed, by measuring some of the stars.

22. On finding the altitude of some of the stars from the hours of the night that have passed.

23. On finding the azimuth of a star from its altitude above the horizon.

24. On finding the distance of a star from the equator, and the parts of the ecliptic that culminate with it, from the azimuth of the place on the horizon circle where it rises or sets. By this the declination of a degree of the ecliptic from the equator is also known.[7]

25. On finding the degree of the ecliptic in which a star is, and the latitude of the star, from its distance from the equator and the degree that culminates with it, when these are known.

26. On finding the distances between the stars according to the arrangement of their places on the sphere in longitude and latitude.

27. On finding the length of the times of the solar year as found by observation, and from it the mean motion of the sun in days, months and years.

28. On finding the irregularity of the motion of the sun, and what appears with it of the degrees of the signs from the places of its farthest distance.[8]

29. On finding the amounts of the difference between days with their nights, when the daytime of one day together with its night is compared with the daytime of another day together with its night, and how they are converted and transferred from one to the other.

[5] Nallino notes that Plato of Tivoli's copy seems to have had «degree» (*daraja*) for «portion» (*wajh*), which may be the better reading.

[6] «The half» is supplied by Nallino; the codex and Plato have «the arc of the day».

[7] Al-Battānī distinguishes the distance of a star from the equator, when the star lies off the ecliptic, from the declination (*mayl*) of the degrees of the ecliptic and of the sun; so the last sentence of the title does not repeat the first.

[8] The Arabic is as ambiguous as Plato's Latin. Nallino takes the sense to be the longitude of the sun reckoned from its apogee, as it comes out from the irregular motion of the sun.

30. On the description of the spheres of the moon and its motions; the irregularity that appears in them at the times of the conjunctions and oppositions with the sun, and the second irregularity that is compounded with it on account of the moon's distances from the sun; the causes of the two kinds of eclipse; the distance of the two luminaries from the earth; and the waxing and waning of the moon's light with its distance from the sun. Ar. p. 4

31. On the description of the spheres of the wandering stars and their states.

32. On finding the eras of the Arabs, the Rūm, the Persians and the Copts, and finding each of them from another.

33. On finding the place of the ecliptic in which the sun is seen, by the era of the Rūm or of the Arabs, whichever of the two you wish.

34. On finding the hours of computation in every locality. These are the mean equal hours after noon at the city of al-Raqqa, by which the motions are found at any time, so that the mean position of a star is known at that time of the day or the night; and on converting these hours into the hours of the localities.

35. On setting up the ascendant and the twelve houses from the hours of the day and the night, and finding the hours from the ascendant.

36. On finding the true place of the moon on the ecliptic on any day and at any time.

37. On finding the place of the northern and the southern node, which are the head and the tail, through which the moon passes in latitude.

38. On finding the latitude of the moon, which is its distance from the belt of the signs towards the south and the north.

39. On finding the parallax that the moon undergoes in longitude and latitude, its amounts in the quarters of the horizon, and the cause from which it arises; in various ways.

40. On finding the distance of the moon from the earth from its parallax in the circle of altitude that lies between the zenith and the horizon and passes through the place of the moon on the ecliptic.

41. On the sighting of the crescent at the beginning and the end of the months; the azimuth of the place in the sky where it is seen; its altitude above the horizon at that time; and its form, according to the light that is in it and to whether its two horns are level or tilted.

42. On finding the mean and the true conjunctions and oppositions of the sun and the moon, by the era of the Rūm or of the Copts, whichever you wish.

43. On finding the lunar eclipses, their amounts and their times in the localities; the side of the moon's disc on which the eclipse begins and the side on which the clearing takes place; and its figure, and working it out by computation and by the tables.

44. On finding the eclipse of the sun, its different amounts in every locality and its times there; the side of the sun's disc on which the eclipse begins and the side on which it clears; and its figure, and working it out by computation and by the tables. Ar. p. 5

45. On finding the places of the five wandering stars on the ecliptic at any time.

46. On finding the stations of the five wandering stars and their retrogradation.

47. On finding the latitudes of the five wandering stars and their directions.

48. On finding the appearance and the disappearance of the five wandering stars.

49. On finding the nine configurations that the fixed stars and some of the wandering stars have with respect to the sun.

50. An account of the distances of the stars from the earth, their diameters, the sizes of their bodies and the extent of their spheres.

51. On finding the motion of the other stars by observation, and on entering in the table the places of those that are needed, in longitude and latitude.[9]

52. On what the masters of talismans have said, that the sphere has a motion of displacement going forwards and backwards, and the error that appears in this.[10]

53. On finding the times of the revolutions of the years, which take place when the sun returns to the place in which it was at the beginning.

54. On determining the amounts of the applications according to the latitudes of the stars, when they cast their rays on the ecliptic.[11]

55. On finding the ascensions of the signs between the quadrants of the sphere.

56. On making the vertical and the horizontal sundial for finding the seasonal hours of the day in every locality; on setting it up correctly and on the direction of the south; and how the direction of the qibla, which is the direction of Mecca the protected, is found on the sundial.[12]

57. The conclusion of the book, and a description of the making of the instrument in the shape of the sphere, which is called «the egg», and of the making of the two instruments for observation.[13]

This is the account of the contents of the book in detail, which comprises fifty-seven sections. Praise be to God for His help, and God bless Muḥammad.

Chapter 1. The opening of the book

Ar. p. 6

He said: The first thing with which every matter is begun and every discourse is opened is the praise of God, exalted be His mention, thanks to Him for His benefits, and the blessing on the Seal of His messengers and prophets — upon them be peace, and the mercy of God and His blessings.[14]

Praise be to God, who created the creatures by His power, ordered all things by His will and perfected them by His wisdom; who «encompasses all things in knowledge» and «has counted all things in number»; from whom «not the weight of an atom in the heavens or on the earth escapes, nor anything smaller than that or greater, but it is in a clear book».[15]

I bear witness that there is no god but God alone, who has no partner, and I bear witness that Muḥammad is His servant and His messenger, whom «He sent with the guidance and the religion of truth, to make it prevail over all religion, though the idolaters dislike it». Through him He guided the believers and «cut off the last of the unbelievers», and He made him a proof for all the worlds. God bless him and his good family, his chosen companions and those who follow his practice, until the Day of Judgement.[16]

To proceed. After the knowledge of the laws and the practice of religion, of which no one may

[9] Where this list has «the other stars», the heading of the chapter itself and Plato's Latin have «the fixed stars».

[10] This is the theory later known as trepidation, an oscillation of the equinoxes. Nallino translates «the masters of talismans» (*aṣḥāb al-ṭilasmāt*) as «the astrologers».

[11] The applications (*ittiṣālāt*) are the astrological aspects between stars.

[12] The sundial is *al-rukhāma*, literally «the marble slab».

[13] Nallino suspects «making» (*ṣanʿa*) where the codex has *ḍaʿa*, «placing»; Plato has «making». The translation follows him.

[14] The Seal of the messengers and prophets is Muḥammad (cf. Qurʾān 33:40).

[15] Qurʾān 65:12, 72:28 and 34:3. The «clear book» is the book that is with God.

[16] Qurʾān 9:33 and 61:9 (Nallino's note has XLI, 9, a misprint for LXI, 9); for «cut off the last of the unbelievers», cf. 7:72 and 8:7.

be ignorant, the science of the art of the stars is among the sciences that are highest in rank and most exalted in standing, finest in adornment, that cling most to the heart and shine most in the soul, and that do most to sharpen thought and reflection, to quicken understanding and to train the mind. For there is a great share of good in it and great benefit: the knowledge of the length of the years and months, of the appointed times, of the seasons, of the lengthening and shortening of the day and the night, of the places of the two luminaries and their eclipses, of the courses of the stars in their direct and retrograde motion, of the changes in their configurations, of the order of their spheres and of all their other relations; and beyond this, what is attained through it by one who examines it closely and reflects on it constantly: the proof of the unity of God, and the knowledge of the essence of the Creator's greatness, of the breadth of His wisdom, the majesty of His power and the subtlety of His handiwork.[17]

Ar. p. 7

He has said — mighty is the Speaker: «In the creation of the heavens and the earth, and in the alternation of night and day, there are signs for those who have understanding.» He, blessed and exalted, has said: «Blessed is He who has placed signs in the sky.» He, mighty and majestic, has said: «It is He who has made the night and the day to follow one another.» He, glory be to Him, has said: «It is He who has made the sun a radiance and the moon a light, and has measured out its stations, so that you may know the number of the years and the reckoning.» And He, exalted be His mention, has said: «The sun and the moon run by a reckoning.» There is much more of this kind in the book of God, mighty and majestic, which it would take long to describe and much space to mention and cite.[18]

When I had studied this science at length and reflected on it constantly, I came to know how the books composed on the motions of the stars disagree, and into what errors some of their authors fell in the procedures they set down in them as foundations and in what they built on these. I also came to know what has accumulated in the motions of the stars over long periods of time, when their observations are compared with the ancient observations; that the inclination of the ecliptic to the equator has been found to be smaller; and what has changed with this change in the various kinds of computation, in the lengths of the years, in the times of the seasons, and in the conjunctions of the two luminaries that are inferred from the times and hours of the eclipses. So I set about correcting all this and making it exact according to the method of Ptolemy in the book known as the Almagest, after close study, long reflection and consideration, following in his footsteps and keeping to what he laid down. For he had investigated these matters from every side and had shown their causes and the reasons that occur in them by a geometrical and numerical proof whose soundness cannot be rejected and whose truth cannot be doubted. After that he ordered that they be tested and examined after him, and said that his observations might be improved upon in the course of time, as he himself had improved upon Hipparchus and others of his peers; for the art is a lofty one, and, being heavenly and vast, it can be grasped only approximately.[19]

I have composed a book on this, in which I have made clear what was obscure, opened what was closed and explained what was difficult in the principles of this science and out of the way in its branches. With it I have made the path of guidance easy for those who follow it and work by it in the art of the stars. In it I have corrected the motions of the stars and their places on the belt of the ecliptic, as I found them by observation and by the computation of the two kinds of eclipse,

[17] The configurations of the stars are the nine configurations with the sun that chapter 49 explains (Nallino).

[18] Qur'ān 3:190, 25:61, 25:62, 10:5 and 55:5. The signs (*burūj*) of 25:61 are the word that al-Battānī uses for the signs of the zodiac throughout the book. In the third quotation the text prints «the night and the moon a light, in succession», a slip that mixes this verse with the next quotation; the Qur'ān (25:62) has «the night and the day», which Nallino translates and the translation follows.

[19] «Has been found to be smaller» renders *mā wujida … min al-taqārub*, literally «the drawing together that was found»: al-Battānī's observations gave a smaller obliquity of the ecliptic than the Greek value (chapter 4). Nallino stresses that the words do not show knowledge of a continuous decrease of the obliquity. For Ptolemy's remark see the Almagest, III 1 (in Heiberg's numbering; Nallino cites Halma's III, 2).

together with all the other procedures that are needed, and I have added other things that are needed. I have arranged that the motions of the stars are found from the tables for noon of the day for which the computation is made, at the city of al-Raqqa, where the observation and the testing that establish all this took place — if God, exalted is He, wills. Success comes from God.

Chapter 2. On the division of the circle of the sphere, and on multiplication, roots and division

Ar. p. 8

He said: The ancients divided the circle of the sphere into 360 parts, and they gave more than one reason for this. One is that the number of these parts is close to the number of days of the year, which is completed when the sun passes a fixed point of the sphere until it returns to it. Another is that it is a number that has a half, a third, a quarter and other fractions that do not come out whole for many numbers. They placed the sun at four points of the sphere, which bring about two equinoxes and two solstices and divide the year into four distinct parts, spring, summer, autumn and winter; and they named each of these points after the season that begins when the sun passes it.

Since everything that has extension has a middle and two ends, each of these seasons divides into three parts, and so the circle of the sphere had to have twelve parts. They found the spring point the best of these points and the most fitting to begin with. For from it, after the equinox, the day begins to lengthen, and the sun is rising into the northern half of its sphere, so that the heat grows; and the nature of this season is moist and inclines to heat, which suits the beginning of growth and the coming into being of things. So they made the reckoning of the sphere begin from it.[20]

Then they found that the constellations next to these twelve parts, which are called signs, were twelve constellations, and they named each sign after the constellation next to it, although over a long time these constellations move away from the places of the signs named after them. So the first part became Aries, followed by Taurus, Gemini, Cancer, Leo, Virgo, Libra, Scorpio, Sagittarius, Capricorn, Aquarius and Pisces.[21]

Each of these signs therefore has 30 parts as its share of the 360 parts of the circle of the sphere. These parts are also called degrees. Each degree is divided into 60 parts, called minutes; each minute is likewise divided into 60 parts, called seconds; each second is divided into 60 thirds; and so on, by the same rule of division, to tenths and the separate orders that come after them.

Ar. p. 9

As for the meaning of multiplication: it is repeating one of two numbers as many times as there are units in the other — I mean the multiplication of units by units. Multiplying fractions by units is repeating the fractions as many times as there are units, or taking of the units the part that the fractions are of one. Multiplying fractions by fractions is taking of one of the two fractions, whichever you wish, the part that the other fraction is of one.

Thus, when degrees are multiplied by degrees, the product is degrees; when they are multiplied by minutes, it is minutes; when they are multiplied by seconds, the product is seconds; and so with what is multiplied by thirds, fourths and the following orders: the product is of the order of the smaller one, by which it was multiplied. As for the orders below degrees, minutes and the others: when each of them is multiplied by itself, the product falls below it by as much as it falls below degrees. For example, minutes multiplied by minutes give seconds, and multiplied by seconds they give thirds, and what is multiplied by thirds and fourths falls by the same rule. Seconds

[20] Nallino notes the same account of the spring point in Abū Maʿshar (d. 886), *Introductorium* II 5, and, almost word for word, in Abraham bar Ḥiyya's *Sphaera mundi* (1165), ch. 2.

[21] The Arabic names are al-Ḥamal (the Ram), al-Thawr (the Bull), al-Jawzāʾ (Gemini), al-Saraṭān (the Crab), al-Asad (the Lion), al-Sunbula (the Ear of Grain), al-Mīzān (the Balance), al-ʿAqrab (the Scorpion), al-Qaws (the Bow), al-Jady (the Kid), al-Dalw (the Bucket) and al-Ḥūt (the Fish). The translation uses the Latin names by which the same signs are known in English.

multiplied by seconds give fourths, and multiplied by thirds they give fifths; and whatever lies further from these orders goes the same way, by the same rule.

When any number of one of these orders, obtained by multiplication or by addition, is divided by sixty, the number at which every order ends and to which all the fractions are related, the result belongs to the next higher order. When, in a number made up of two or more of these orders, more has to be subtracted from one of them than it contains, one unit is broken off from the next higher order for it, counted as 60 parts and added to it; then as much as needed is subtracted, and what remains is counted together with what remains of the higher order.

As for degrees: whatever accumulates of them when the parts of the motions are added is reckoned in revolutions. If the sum is more than one revolution, or than several revolutions, a revolution being 360 degrees, the revolutions are dropped and what remains is counted. When more has to be subtracted from the degrees than they contain, a revolution is added to them; then as much as needed is subtracted from the sum, and what remains is counted.

Ar. p. 10

If you wish to multiply one of the orders of degrees or of fractions by another of them, and to know from this table of which order the product will be, take on one of the two lines AB the cell in which the order you wish to multiply by any other order is written, and go straight on from that cell until you are level with the other order that you want: the cell level with it, among the orders of fractions, gives the order of the product. For example, suppose you wish to multiply fourths by thirds. Take on the line AB that runs across the width of the page either of the two orders — let it be the thirds first — and go on from it until you are level with the fourths on the line AB that runs along the length of the page; in the cell level with it you find sevenths, and this is the order of the product. Likewise, if you took the fourths on a line AB and went from them to the level of the thirds on the other line AB, you would find sevenths there. Do the same with any orders you wish, if God wills.

The orders of products[22]

degrees	minutes	seconds	thirds	fourths	fifths	sixths	sevenths	eighths	ninths	tenths
minutes	seconds	thirds	fourths	fifths	sixths	sevenths	eighths	ninths	tenths	elevenths
seconds	thirds	fourths	fifths	sixths	sevenths	eighths	ninths	tenths	elevenths	twelfths
thirds	fourths	fifths	sixths	sevenths	eighths	ninths	tenths	elevenths	twelfths	thirteenths
fourths	fifths	sixths	sevenths	eighths	ninths	tenths	elevenths	twelfths	thirteenths	fourteenths
fifths	sixths	sevenths	eighths	ninths	tenths	elevenths	twelfths	thirteenths	fourteenths	fifteenths
sixths	sevenths	eighths	ninths	tenths	elevenths	twelfths	thirteenths	fourteenths	fifteenths	sixteenths
sevenths	eighths	ninths	tenths	elevenths	twelfths	thirteenths	fourteenths	fifteenths	sixteenths	seventeenths
eighths	ninths	tenths	elevenths	twelfths	thirteenths	fourteenths	fifteenths	sixteenths	seventeenths	eighteenths
ninths	tenths	elevenths	twelfths	thirteenths	fourteenths	fifteenths	sixteenths	seventeenths	eighteenths	nineteenths
tenths	elevenths	twelfths	thirteenths	fourteenths	fifteenths	sixteenths	seventeenths	eighteenths	nineteenths	twentieths

[22] The print reads from right to left; here the columns are set from left to right. The first row is the line AB across the width of the page, the first column the line AB along its length. The cell of the fourth row and second column is printed «ورابع», a misprint for «روابع», «fourths».

Ar. p. 11

As for the meaning of the root: the root of any absolute number, whatever number it is, is that which, multiplied by itself, gives the given number. But the roots of these orders do not keep this rule, because of the difference that we have described in what results from multiplying these orders by one another. The rule holds only for the order of degrees: the root of degrees is also degrees, since degrees multiplied by degrees give degrees. As for the fractions below degrees, of the remaining orders: those of even rank, such as seconds, fourths, sixths and the like, have their root in the higher order whose rank is half of theirs — seconds, whose root is minutes, and fourths, whose root is seconds. Those of odd rank, such as minutes, thirds and the like, have no definite root unless they are converted into the order below them, so that they come to an even rank; then this rule applies to them: minutes are converted into seconds, and thirds into fourths.[23,24]

Division is finding how many times the smaller number goes into the larger, when the larger is measured by the smaller, and finding what part of the larger the smaller is, when the smaller is the number divided. Here we proceed in the reverse of what we did in multiplications and roots, by the same rule: if we divide degrees by degrees, the quotient is degrees. As for the other orders below degrees: when a lower order is divided by a higher one, whatever its rank and whether or not it is next to it, the quotient is of the order which, multiplied by the order divided by, gives back the order of the number divided. Thus seconds divided by minutes give minutes, and sixths divided by fourths give seconds. But when a higher order is divided by a lower one, the way to do it is to convert the higher order into the lower and then divide by it, and the quotient is degrees. So with minutes divided by sixths: when they are converted into sixths and then divided by those sixths, the result of the division is degrees, as we have described.

Ar. p. 12

If you wish to know, with the table given here, what results from dividing the lower orders of fractions by the orders above them, look up in the line AB or in the line AD, whichever you wish, the order that you wish to divide by an order of higher rank, whether or not it is next to it, and go on level with it until you are level with the higher order on the other line. The order of fractions at which you arrive is the order that you obtain by dividing the number divided, and it is the order which, multiplied by the higher order that you divided by, gives back the order of the number divided. Likewise, if you wish to divide a higher order by a lower one, convert the higher into the lower, look up in one of the two lines the order into which it has been converted, and go on level with it until you are level with the order that you wish to divide by: you obtain degrees. And whenever you divide an order by the same order, the result is degrees, if God, exalted is He, wills. Success comes from God.[25]

[23] An absolute number (*ʿadad muṭlaq*) is a number taken by itself, not counted in one of the sexagesimal orders.

[24] The print has «فتلزَمه» (p. 11, line 7), a misprint for «فتلزَمه», «applies to them», which the corrigenda do not list.

[25] «With the table given here»: the Arabic says «with this table mentioned before»; the table is printed at the head of the next page.

The orders of quotients[26]

degrees	minutes	seconds	thirds	fourths	fifths	sixths	sevenths	eighths	ninths	tenths
minutes	degrees	minutes	seconds	thirds	fourths	fifths	sixths	sevenths	eighths	ninths
seconds	minutes	degrees	minutes	seconds	thirds	fourths	fifths	sixths	sevenths	eighths
thirds	seconds	minutes	degrees	minutes	seconds	thirds	fourths	fifths	sixths	sevenths
fourths	thirds	seconds	minutes	degrees	minutes	seconds	thirds	fourths	fifths	sixths
fifths	fourths	thirds	seconds	minutes	degrees	minutes	seconds	thirds	fourths	fifths
sixths	fifths	fourths	thirds	seconds	minutes	degrees	minutes	seconds	thirds	fourths
sevenths	sixths	fifths	fourths	thirds	seconds	minutes	degrees	minutes	seconds	thirds
eighths	sevenths	sixths	fifths	fourths	thirds	seconds	minutes	degrees	minutes	seconds
ninths	eighths	sevenths	sixths	fifths	fourths	thirds	seconds	minutes	degrees	minutes
tenths	ninths	eighths	sevenths	sixths	fifths	fourths	thirds	seconds	minutes	degrees

[26]The columns are set from left to right, as in the first table. The first row is the line AB, the first column the line AD. The cells of the second row and fourth column and of the fourth row and second column are printed «ثولنٍ» and «توانٍ», misprints for «ثوانٍ», «seconds».

Chapter 3. On finding the lengths of the chords of the parts of the circle, on entering in tables the halves of the chords of double the arcs, and on all the procedures with them that follow from this[27]

Ar. p. 13

He said: The ancients disagreed about the size of the diameter of the circle in relation to its circumference, although they came close to it. Some said that the circumference of the circle is 3 times its diameter and 1/7 of it. Others said that it is 3 times [the diameter] and a little more than 10 parts of 71. What the excellent Ptolemy and the astronomers worked with lies between these two amounts: 3 times [the diameter] together with 1/10 of it and 1/4 of 1/6 of it. We do not need to know the truth of this in setting out the chords, since arcs and chords have no known measure in relation to one another, and this is known only from their chords. Nor does it do us any harm to take the diameter as large as we wish. This is why Ptolemy made it 120 parts, because the computation comes out easily this way; and we also work with it in this book.[28]

Ar. p. 14

It has been shown by proof that the chord of a sixth of any circle is equal to its radius. A sixth of the circle of the sphere is evidently 60 parts, according to the basis of the computation, in the measure in which the circle is 360 parts; and the chord of the sixth is also 60 parts, in the measure in which the diameter is 120. If the chord of a sixth of the circle is multiplied by itself and subtracted from the whole diameter multiplied by itself, the root of the remainder is the chord of a third of the circle. In the same way, for any arc of a circle whose chord is known: if the chord of that arc is multiplied by itself, the product is subtracted from the whole diameter multiplied by itself, and the root of the remainder is taken, the result is the chord of the arc that remains to complete the semicircle. The chord of a quarter of the circle is the root of twice the product of its radius by itself. The chord of a tenth of any circle is found thus: to the radius multiplied by itself add a quarter of the diameter multiplied by itself, take the root of the sum, and subtract from it a quarter of the diameter of the circle; what remains is the chord of a tenth of that circle. The chord of a fifth of the circle is found thus: multiply the chord of its tenth by itself, add to it the radius multiplied by itself, and take the root of the sum; this is [the chord of] the fifth of that circle.

For any two arcs of a circle whose chords are known, the chord of the arc by which they differ is also known. Multiply the chord of each of the two arcs by the chord of what remains of the other to complete the semicircle, take the difference between [the two products] and divide it by the diameter; the result is the chord of the arc by which the two arcs differ.

For any arc of a circle whose chord is known, the chord of its half is also known. Subtract the chord of what remains of that arc to complete the semicircle from the whole diameter of the circle, take half of the remainder, multiply it by the whole diameter, and take the root of the product; the result is the chord of half that arc.

For any two arcs of a circle whose chords are known, when one is placed on the other and we join them so that they become one arc, the chord of the combined arc is also known. Multiply the chord of each of the two arcs by [that of] the other, and the chord of what remains of each of them to complete the semicircle by [that of] the other as well; take the difference between [the two products] and divide it by the whole diameter. The result is the chord of what remains of the combined arc to complete the semicircle. If it is multiplied by itself and subtracted from the whole diameter multiplied by itself, the root of the remainder is the chord of the arc combined of

[27] The translation follows Nallino's corrigenda in three places: «لتمام» for «لتمام» (p. 14, line 4), «وتر» for «وثر» (p. 14, line 18) and «حاجتنا» for «جاجتنا» (p. 15, line 19). It also reads «نحتاج» for «نحناج» (p. 15, line 11), «والوتر» for «واوتر» (p. 16, line 14), «يتلوه» for «بتلوه» and «فزدته» for «فردته» (p. 16, line 17) and «الوتر» for «اوتر» (p. 17, line 11), misprints that the corrigenda do not list.

[28] 3 1/7 and 3 10/71 are the two bounds that Archimedes proved for this ratio in the *Measurement of the Circle*. Ptolemy's value, 3 + 1/10 + 1/24 = 3 17/120, is 3;8,30 in sexagesimal fractions (*Almagest* VI 7). The ratio that is known only from the chords is that of the chord of one arc to the chord of another (Nallino).

the two arcs. By this rule and in these ways all the remaining chords in the semicircle that can be known are found.

As for the chords that are not known by proof, such as the chord of one degree and its doublings, two, four, eight and the like: they are not found by computation in the way of proof, as those arcs and those chords are. But it is known by proof that the ratio of the chord of a smaller arc to its arc is greater than the ratio of the chord of a greater arc to its arc. Now the chord of one and a half degrees and the chord of a half and a quarter [of a degree] are known by proof, and 2/3 of the chord of one and a half degrees is equal to the chord of a half and a quarter of a degree increased by a third of itself; there is no perceptible difference between them, and no harm to the computation comes from it. If the chord of three quarters [of a degree] is taken and a third of it is added to it, the sum is the chord of one degree. Once the chord of one degree is known in this way, all the chords of the degrees of the semicircle become known as well.[29]

Ar. p. 15

What is needed about the amounts of arcs that intersect one another in known arcs is found only through the chords of double the known arcs. When a circle is divided by two lines that intersect at its centre at right angles, it is divided into four equal quarters, at four angles; each angle takes in 90 parts of the circumference and is enclosed by two lines that go out from the centre to the circumference, each of them the radius. The two right angles under two quarters together are enclosed by one straight line, the whole diameter, which is plainly twice each of the two lines that enclose the one right angle under one quarter. So the ratio of each of the two lines that enclose a right angle to the diameter that encloses the two right angles is as the ratio of a quarter of the circle to its half. Hence the diameter divides the chords of the remaining arcs in the semicircle into two halves, and it divides the arcs with them into halves on its two sides. So the ratio of the chord of each of these arcs to the whole diameter is as the ratio of half that chord, which lies under half that arc, to the radius; and this is half the chord of twice the arc that lies on either side of the diameter, half of it in each of the two quarters. This is what we mean, and it is what we use in the procedures of computation, so that we do not need to double the arc in what we seek to find. Ptolemy did that only to establish the proof.[30]

We, however, have taken half the chord of twice each arc of the quarter of the circle and entered it under the entry for that arc in the quarter, and we have made the arcs in the tables increase by half a degree up to the full 90 degrees that make up the whole quarter. So half the chord of one degree stands under half a degree; half the chord of 60 under 30 degrees; half the chord of 120 under 60; and half the chord of 180 degrees — the semicircle, whose chord is the whole diameter — under 90, the degrees of the whole quarter: this is the radius, whose amount is 60 parts. All these halved chords entered in this book are related to it. So that we need not repeat ourselves in what follows, we state that whenever we speak of chords in this book, we mean these halved chords, except where we name one specially and call it a full chord; and for the most part we seldom need these.[31]

Ar. p. 16

If you want to find from the tables the sine of any degree you like, look in the table of sines, in the column of the numbers that increase by half a degree; where you find the number that you have, take the degrees, minutes and seconds entered opposite it in the table of sines: that is the sine of the degrees you wanted. If there are minutes with the degrees, more or less than 30 minutes, take what stands opposite the whole degrees, or the degrees and halves, whichever is nearest to the degrees and minutes you have and smaller than them, and keep what you find opposite

[29] This is Ptolemy's procedure for the chord of 1° (*Almagest* I 10). By the inequality just stated, the chord of 1° lies between 2/3 of the chord of 1½° and 4/3 of the chord of ¾°, and these two bounds agree to the precision of the tables.

[30] The first sentence is obscure; Nallino, following Schiaparelli, takes it to mean that every computation that combines several arcs is made through the chords of the doubled arcs.

[31] From here on the translation calls these halved chords «sines», as Nallino does in his Latin; al-Battānī calls them «halved chords» (*awtār munaṣṣafa*) or simply «chords». Other Arabic authors call the sine *jayb*. Since the radius is 60, al-Battānī's sine of an arc is 60 times its sine in modern terms. The full chords are needed in chapters 26, 50 and 54 (Nallino).

it in the table of sines. Then subtract the number you found in the column from the one you have, multiply the minutes that remain by the difference between the sine you kept and the sine opposite the number half a degree greater, and divide the product by the 30 minutes by which the numbers in the two columns of arcs increase. Add the minutes and seconds that result from the division to the sine you kept if it is the smaller, and subtract them from it if it is the greater; what you obtain after the addition or the subtraction is the sine of the degrees and minutes that you have.[32]

Or, if you wish, find what part of 30 minutes the minutes left over are: if they are a half, or a third, or less or more than that, take the same part of the difference of the sines and proceed with it in the same way, adding or subtracting; the result is the sine of the arc you wanted.

If you wish to find arcs from these sines, look for the sine in the table of sines; where you find it, or the nearest value smaller than it, take what stands opposite it in the first of the two columns of numbers: that is the arc you want, and keep it. Then subtract the sine you found in the tables from the sine you have, multiply the remainder by 30 minutes, and divide the product by the difference between the sine you found and the one that follows it. Add the minutes and seconds that result to the arc you kept; the arc you then have is the arc of the sine you want. Or, if you wish, see what part the minutes and seconds left over are of the difference between the sine you found and the one that follows it, take the same part of 30 minutes and add it to the arc you kept, as before; it comes to the same. This procedure is called taking the arc of the sines.[33]

If you wish to find the versed sines from the arcs: if the number whose versed sine you want is less than 90 degrees, subtract it from 90, find the sine of the remainder by the rule given above, and subtract it from 60 degrees, which is the radius; what remains is the versed sine of that arc. If the number is more than 90, find by how much it exceeds 90, find the sine of the excess, and add it to 60 degrees; the sum is the versed sine of the arc you wanted.[34]

Ar. p. 17

If you wish to find the arcs from these versed sines: if the versed sine you have is less than 60 degrees, subtract it from 60, find the arc of the remainder by that rule, and subtract the arc from 90; what remains is the arc of the versed sine. If the versed sine is more than 60, take 60 from it, find the arc of the remainder, and add it to 90 degrees; the sum is the arc of the versed sine.

You need no more for finding arcs and sines than what I have set out for you. For these sines it is enough to know the sines from 1 degree to 90 degrees, because the sine of whatever goes beyond 90, up to 180, is like the sine of the degrees up to 90 taken in reverse order. Likewise for the full chords you need no more than the chords of the semicircle, from 1 degree to 180, because the chords of the other half are like those of the 180 degrees in reverse order.

As for finding the full chords from the arcs, and the arcs from these chords: if you wish to find the full chord of any number of degrees, take half those degrees, find its sine from the table, and double it; the result is the full chord of the degrees you wanted. If you wish to find the arcs of the full chords with the tables as well, take half the chord, find its arc from the table in the way described above, and double the arc; the result is the arc of the full chord that you wanted.

Whenever one of these sines is multiplied by itself and subtracted from the radius multiplied by itself, the root of the remainder is the sine of what remains of that arc to complete a quarter of the circle. And when the sine of any degree is subtracted from the radius, the remainder is multiplied by 30 parts and the root of the product is taken, the result is the sine of half of what remains of that arc to complete a quarter of the circle, if God wills.[35]

[32] The sines are given in degrees, minutes and seconds of a radius of 60 degrees.

[33] «Taking the arc» is *taqwīs*, Nallino's *arcuatio*: finding the arc that belongs to a given chord or sine.

[34] The versed sine is *al-watar al-rāji*ʿ, literally «the returning chord». In modern terms the versed sine of an arc a is $60(1 - \cos a)$; the two rules of the text are the cases $a < 90°$ and $a > 90°$.

[35] In modern terms, writing Sin a = 60 sin a for al-Battānī's sine of an arc a: $\text{Sin}(90° - a) = \sqrt{(60^2 - (\text{Sin } a)^2)}$, and $\text{Sin } ½(90° - a) = \sqrt{(30\ (60 - \text{Sin } a))}$, because $60^2 \sin^2 ½(90° - a) = 1800\ (1 - \cos(90° - a)) = 30\ (60 - \text{Sin } a)$.

Chapter 4. On finding the amount of the inclination of the ecliptic to the equator, the division of this inclination, its directions, and its ranks in its ascent and descent; this is the declination of the sun from the right sphere

Ar. p. 17
Ar. p. 18

He said: The ecliptic is defined by the path on which the sun is seen to move; the equator is the circle along which the greatest sphere turns about its two poles. The inclination of the ecliptic to the equator is found only by observing the sun and following its passage through the two solstitial points on the meridian circle. This is the circle of the middle of the sky; it passes through the two poles of the equator and the zenith, and it cuts the horizon circle.

Hipparchus stated, and Ptolemy reports in his book, that the arc of the meridian circle between the winter and the summer solstice is 47° 42′, and that the inclination is half of that, 23° 51′. We ourselves observed it many times in our own day with the long alidade and with the brick, whose making and setting up are described in the Almagest, after dividing [their scales] precisely and setting up the instrument as exactly as could be done. We found the least distance of the sun from the zenith on the meridian circle, in the city of al-Raqqa, to be 12° 26′, and its greatest distance 59° 36′. From this it became clear to us that the arc between the two solstices is in truth 47° 10′, and that the inclination of the ecliptic to the equator is half of these degrees, 23° 35′; this is the distance between the poles of the two circles. We work with this value in this book, since we have seen it with our own eyes, while the first is known by report.[36,37]

From this we also learned that the distance of the city of al-Raqqa, where the observations were made, from the equator on the meridian circle is 36°. This is the altitude of the north pole of the equator above the horizon there, and it is also the distance of the equator from the zenith towards the south.[38]

If you wish to compute this inclination, so as to find the part of it that belongs to each degree you want, up to the full 90 degrees that complete the inclination, which is 23° 35′: take the sine of one degree, or two degrees, or more, up to the full 90 degrees, which run from the beginning of Aries to the end of Gemini. When you know the sine of the degrees you wanted, multiply it by the sine of the whole inclination, divide the product by the radius, which is 60, and find the arc of the quotient. The arc you obtain is the declination of the degree you wanted from the equator, on the meridian circle.[39]

Ar. p. 19

If you wish to enter this in a table degree by degree, find the declination of each degree you want up to 90 degrees, and enter under each degree what belongs to it. When you have done this, you know the declination of all the parts of the ecliptic. For the declination of what goes beyond 90, up to 180, is like the declination of the 90 [degrees] entered, taken in reverse order; the declination of what goes beyond 180, up to 270, is like the declination of the 90 in the same order; and the declination of what goes beyond 270, up to 360, is like the declination of the 90 in reverse order. We have entered this in the table at intervals of one degree, with four columns of numbers. In the first column we have entered the numbers of the ecliptic from 1 degree to 90; in the second, what the numbers of the first column fall short of 180; in the third, the numbers of the first column added to 180; and in the fourth, what the numbers of the first column fall short

[36] Ptolemy (*Almagest* I 12) found the arc between the solstices to be more than 47 2/3° and less than 47 3/4°, nearly the ratio of 11 to 83 parts of the meridian that he ascribes to Eratosthenes and says Hipparchus also used; his table of declinations (*Almagest* I 15) uses an inclination of 23° 51′ 20″.

[37] The long alidade (*al-ʿiḍāda al-ṭawīla*) is Ptolemy's parallactic instrument, the triquetrum (*Almagest* V 12); the brick (*al-libna*) is his plinth, a quadrant set up in the plane of the meridian (*Almagest* I 12). Al-Battānī describes both in chapter 57 (Nallino).

[38] The two observations give 36° 1′, as Delambre noted, since the equator lies midway between the two solstitial points: (12° 26′ + 59° 36′)/2 = 36° 1′. Nallino suggests that al-Battānī dropped the minute to keep his tables simple, 36° being also the latitude of Rhodes, for which Ptolemy computed several of his tables.

[39] In modern terms, $\sin \delta = \sin \lambda \cdot \sin \varepsilon$, where λ is the distance of the degree from the beginning of Aries, ε the inclination and δ the declination.

of 360. Thus, if the number falls in the first or the second column, we know that the declination is towards the north of the equator, and if it falls in the other two columns, we know that it is towards the south.

If you wish to find the declination of the sun, or of any other degree of the signs, take [the distance] from the beginning of Aries to the degree of the sun, or to whatever other degree whose declination you wish to find; the result is the argument of the declination. Look for the same number in the four columns of numbers entered in the tables of declination, and take the degrees, minutes and seconds entered opposite it: that is the declination of the degrees you wanted. If there are minutes with the degrees, take for them their share of the difference, in the way I showed you for the differences of the sines. See what part the minutes are of the 60 minutes by which the numbers increase, and take the same part of the difference between the declination of the whole degrees and the declination of the degree one greater. Then look: if the declination of the degree you have is the smaller, add what you obtained to it; if it is the greater, subtract it from it. The declination you obtain after the addition or the subtraction is the declination of the degrees and minutes you wanted from the equator.[40]

If the argument is from 0 to 90, the declination is increasing and the sun is ascending in the north; if it is from 90 to 180, the declination is decreasing and the sun is descending from the north; if from 180 to 270, the declination is increasing and the sun is descending towards the south; and if from 270 to 360, the declination is decreasing and the sun is ascending in the south. In short, when the argument of the declination is from 0 to 180, the declination is northern, and when it is from 180 to 360, it is southern. By this computation you find the declination of the sun, its direction, and its ascent and descent.[41]

They have divided the declination into six ranks of ascent and descent: they made every 15 degrees of the sun's course in each of these quarters one of the ranks of ascent and descent, up to the full 90 degrees that complete the six ranks. When the sun is in the first 15 degrees of one of the quarters, they say that it is in the first rank; when it is in the second 15 degrees, they say that it is in the second rank; and so on until the sixth rank is reached.[42]

Chapter 5. On finding the ascensions of the signs in the right sphere[43]

Ar. p. 20

He said: Suppose you wish to know how many of the 360 times of the equator rise together with given parts of the ecliptic. These are the ascensions of the signs at the place of the line of equality. This is the place that has no latitude, above which the equator turns; there night and day are always equal, on all the days of the year. In every locality the signs cross the middle of the sky with the same amount [of the equator] with which they rise on this line, and there they also cross the middle of the sky with this amount. This is why these are called the ascensions of the signs in the right sphere. Every three signs rise in the right sphere together with 90 times of the equator.[44]

If you wish to compute the ascension in the right sphere of any degree of the signs you like, take the whole inclination, which is 23° 35′, and find its sine: this is the sine of the whole inclination. Then subtract the whole inclination from 90 and find the sine of the remainder: this is the sine of the complement of the whole inclination. Then take [the distance] from the beginning of Aries

[40] «The argument of the declination» is *ḥiṣṣat al-mayl*, literally «the share of the declination»: the number with which the table is entered (Nallino's *argumentum declinationis*).

[41] The text writes these numbers in abjad numerals, with the letters of the alphabet, and 0 with the zero sign of the codex.

[42] Theon of Alexandria describes the same ranks (Greek *bathmoi*) of 15° as a usage of the astrologers; al-Battānī gives a table of them at the end of his table of the declination of the sun (Nallino).

[43] The translation reads «تلك», «that», for the misprint «تلك» (p. 20, line 14), which the corrigenda do not list.

[44] A time (*zamān*) is a degree of the equator, the measure of the daily turning of the sphere. The three signs are those of a quarter of the ecliptic that begins at an equinoctial or a solstitial point.

to the degree you want, find the declination of that degree, and find its sine: this is the sine of the declination of the degree. Then subtract the declination of the degree from 90 and find the sine of the remainder: this is the sine of the complement of the declination of the degree. Then multiply the sine of the declination of the degree by the sine of the complement of the whole inclination, divide the product by the sine of the whole inclination, multiply the quotient by the radius, which is 60, divide the product by the sine of the complement of the declination of the degree, and find the arc of the quotient. The arc you obtain is the amount of the equator that rises from the beginning of Aries to the degree you wanted.[45]

If you computed for 30 degrees, the result is the ascension of the whole sign of Aries; if you computed for 60, it is the ascension of Aries and Taurus together: subtract from it the ascension of Aries, and the ascension of Taurus remains. Then subtract the ascension of Aries and Taurus together from 90; what remains is the ascension of Gemini. When you know the ascension of Aries, the ascensions of Virgo, Pisces and Libra are equal to it; the ascensions of Leo, Aquarius and Scorpio are equal to that of Taurus; and the ascensions of Sagittarius, Capricorn and Cancer are equal to that of Gemini.

Ar. p. 21

By this rule you derive the ascensions degree by degree and enter them in the tables. Begin from the beginning of Capricorn, so that from it you know the ascensions of the signs in the middle of the sky for every locality, and so that the number by which one finds what culminates and what parts rise in every locality is one and the same number. We shall explain how the ascensions are put into tables when we come to the ascensions of the signs in the climates, so as not to repeat ourselves.[46]

Chapter 6. On the properties of the parallel lines that run parallel to the equator, on the places of the inhabited earth that are known in longitude and latitude, and on what follows from this[47]

Ar. p. 21

He said: We should begin with the equator, and then speak of the other circles that decline from it towards the north, and of the places of the earth that lie beneath these circles.

We say: The line of the earth that lies beneath the equator is the line called the line of equality. It has no latitude; the equator turns upon it and above it, from east to west, and there day and night are always equal, on all the days of the year, as we said at the beginning. When the sun passes over this line, and only then, day and night are balanced and perceptibly equal over the whole earth. The sun is then at the point common to the ecliptic and the equator, for this is where the two circles intersect: the point of the beginning [of Aries and the beginning] of Libra. Only then does the sun stand at noon above the heads of those who live on this line, so that upright objects there cast no shadow. When the sun runs in the northern half of the ecliptic, the shadows of upright objects there fall towards the south at noon; when it runs in the southern half of the ecliptic, the shadows then fall towards the north at noon. This line alone bounds the whole inhabited quarter [of the earth] on the south. Also, all the stars rise and set there, because the two poles of the sphere lie there on the horizon circle itself, and the turning of the sphere

[45] In modern terms, $\sin \alpha = \tan \delta \cdot \cot \varepsilon$, where α is the ascension of the degree in the right sphere, δ its declination and ε the inclination.

[46] Counted from the beginning of Capricorn, the ascension in the right sphere of the culminating degree is equal to the ascension at the locality of the rising degree, counted from the beginning of Aries; one number therefore gives both (Schiaparelli, in Nallino's commentary). Al-Battānī's table of ascensions in the right sphere is counted in this way. The ascensions in the climates are the subject of chapter 13.

[47] The translation reads «الشمس», «the sun», for «اشمس» (p. 23, line 18) and «تنتهي» for «ننتهي» (p. 24, line 22), misprints that the corrigenda do not list. It follows the emendanda printed in Part II in reading «الأندلس» for «الأندلس» (p. 26, line 18) and «اوقيانس» for «ارقيانس» (p. 28, line 12).

there is like that of a water-wheel, the wheel of a well.[48,49]

Ar. p. 22

It is not known for certain that this line of the earth is inhabited, for we have seen no one who claims to have reached it in our time, and Ptolemy does not mention it in his book. But people of understanding know that the climate of this line is temperate, because the sun never goes very far from it and does not stay long over its zenith, since at that time it passes quickly through the declination. So summer and winter there are both mild. Something like what we have described is seen in places near it, such as Ṣanʿāʾ, Aden and the other places of Yemen that are close to it.[50]

As for the other lines that decline from this line towards the north, which are the lines parallel to it: on any one of them, the stars that lie within the circle drawn about the north pole of the equator with a radius equal to the altitude of the pole above the horizon on that line never set on that line. Of the stars outside this circle, those that are near it may be seen twice in one night, once at the beginning of the night and again at its end, and they set in between. The stars at the zenith of one of these lines are those that cut off, on the circle through the two poles of the equator, an arc between the star and the equator equal to [the arc] between the line and the equator. The stars that are always hidden are those within the circle drawn about the south pole of the equator with a radius equal to the depression of that pole below the horizon. All the other stars, which are not of this description, rise and set.

Therefore, if one observes one of the stars within the circle drawn about the north pole of the equator with a radius equal to the altitude of the pole above the horizon, and takes its altitude at its highest, when it crosses the line of the middle of the sky above the pole, between the zenith and the pole; if one then waits until it is on the line of the middle of the sky below the pole, between the pole and the horizon, where it is lowest, and takes its altitude then; and if one finds the difference between the two altitudes and adds half of it to the smaller altitude: the result is the altitude of the north pole above the horizon there. Likewise, if the two altitudes are added together and half the sum is taken, this is the altitude of the pole, and it is the latitude of that place to the north.

Ar. p. 23

On every one of these lines whose distance from the equator is less than the inclination, the sun passes over the zenith twice a year. This is known from the table of declination, and so is the part of the ecliptic where it happens, I mean the degree in which the sun is on that day. For when the sun is at the beginning of Aries or of Libra, it passes at noon over the zenith on the line of equality, as has also been said before; and there this happens only once a year. When it is north of these two points, it reaches the zenith of those who live beneath its path in declination, that is, where the altitude of the pole equals the declination of the degree in which the sun is on that day, the declination being northern. It is known that the distance of the equator from the zenith is then equal to the declination, and that on that day upright objects there cast no shadow at noon. When the sun moves away from them, the shadows of upright objects at noon fall towards the south, until it returns to them and stands over their heads a second time. Then again upright objects have no shadow, until it moves away from them, and the shadows of upright objects fall towards the north.[51]

On the other lines, whose distance from the equator is greater than the inclination, the sun never reaches the zenith of their inhabitants, and the shadows of upright objects there never fall towards the south at noon. On them the difference between night and day in length keeps

[48] The words in brackets are supplied by Nallino; the codex lacks them.

[49] The print has «السَّنْويّ», «annual», which gives no sense here; Nallino's emendanda, printed in Part II, read perhaps «السّانويّ», from *sāniya*, the wheel that lifts water from a well. The sphere of the equator turns like a wheel standing upright, with its axis lying in the horizon.

[50] *Almagest* II 6 (Nallino).

[51] Once a year at each of the two points (Nallino); Plato of Tivoli has «twice», perhaps a conjecture of his own, since the codex has «once».

growing, up to the line whose distance from the equator is 66° 25′, which is what the whole inclination falls short of 90. On this line alone, when the sun reaches the summer solstitial point, which is called the beginning of Cancer, the day is longer by 12 hours, so that day and night together become one day, all of it daylight; and the night becomes the same when the sun reaches the winter solstitial point, which is called the beginning of Capricorn. This line alone is the first of the lines on which the shadows of upright objects fall towards all the directions of the horizon, because beyond this line, towards the north, the middle of the day is not bounded. On this line alone the ecliptic coincides with the horizon itself when the spring equinoctial point rises; for the beginning of Cancer then rises at the north point, and at the same time the beginning of Aries is on the eastern horizon, rising at the equinoctial rising point. Therefore, when the sun passes the summer solstitial point, it does not set for a day and a night: it travels around the horizon at varying distances from it until it returns to the north point, so that this day has no night at all.

Ar. p. 24

He said: On the other lines that decline from this line towards the north, the shadows of upright objects go round them, on every one of these lines, towards all the directions of the horizon, and the length of the day on each of them is known from the table of declination. For when the declination of the degrees at which the sun, on these lines, is distant from the solstitial point is subtracted from 90, the remainder is the distance of the parallel line from the equator, I mean the altitude of the pole on that line. The degrees that the sun traverses on the two sides of the solstitial point are either always visible or always hidden. So on some of these lines the day lasts a month, or two months, or three, or less or more, and the night lasts as long at the opposite season. This goes on up to the line whose distance from the equator is all the degrees of the quarter, where the altitude of the pole is 90 degrees. There the day, with the shadow of upright objects going round them towards all the directions of the horizon, lasts nearly six months. For there the northern half of the ecliptic, from the beginning of Aries to the end of Virgo, is never hidden below the earth, and its other, southern half never appears above the earth; so the whole year is a single day, half of it daylight and half of it night. There alone the north pole is at the zenith, and the equator is at once always visible and always hidden, for it lies in the very place of the horizon there. Clearly the turning of the sphere there is like that of a millstone. Between the line of equality and this line, the turnings vary between that of the water-wheel and that of the millstone: the turning inclines with the inclination of the place from the line of equality, according to its nearness to each of the two lines and its distance from it. God knows best.[52]

We give an example of some lengths of the day on some of these lines, to serve as a model. We take it on the line whose distance from the equator is 69° 44′, which is the altitude of the pole on this line. Subtracting this from 90 leaves 20° 16′. We find that the sun reaches this amount of northern declination when it is 30 degrees from the summer solstitial point on either side; that is, from the time it is at the beginning of Gemini it remains visible above the earth, going round the horizon, until it reaches the beginning of Leo. So on this line it does not set at all while it is in these 60 degrees on the two sides of the solstitial point, and the day, with the shadow of upright objects going round them towards all the directions of the horizon, lasts until the sun has crossed these degrees, which takes about two months. When its whole distance from the winter solstitial point lies between the two degrees opposite these two degrees, it does not appear above the earth, that is, from the time it reaches the beginning of Sagittarius until it reaches the beginning of Aquarius; so the night too lasts about two months.

Ar. p. 25

As for the line whose distance from the equator is 78° 28′: there the sun does not set below the earth when it reaches a northern declination equal to what these degrees fall short of 90, and does not appear above the earth when it reaches a southern declination of the same amount. This amount is 11° 32′. We find that the sun reaches this declination when it comes to the beginning of Taurus and to the beginning of Virgo, where its distance from the summer solstitial point is

[52] Al-Farghānī makes the same comparison with the millstone at the end of his chapter 7, and Dante took it from him into the *Convivio* (Schiaparelli, in Nallino's note).

60 degrees on either side. So the day, with the shadows of upright objects going round them, lasts nearly four months. Likewise the sun stays below the earth while its course runs from the beginning of Scorpio to the beginning of Pisces, so that the night too lasts as long.[53]

As for the known places of the earth and the inhabited localities in longitude and latitude: we have shown, by the reasoning that Ptolemy set out and in which others of the ancients agreed with him, that the earth is round, that its centre is in the middle of the sphere, that the air surrounds it on all sides, and that compared with the ecliptic it is as small as a point.[54]

As for its inhabited part, they took its limits from the inhabited islands called the Eternal Islands, in the western Ocean, which are six inhabited islands, to the farthest inhabited land of China, and they found this to be 12 hours. So they knew that when the sun sets at the farthest inhabited land of China, it begins to rise over the first of the inhabited islands said to be in the western Ocean, and when it sets at these islands, it begins to rise over the farthest inhabited land of China. That is half the circle of the earth, and it is the length of the inhabited land as far as it has been explored. In miles it is 13,500 miles, in the miles that they used in measuring the earth. Then they considered the latitude, and found that the inhabited land runs from the place of the line of equality towards the north as far as the island of Thule, which is in Britain, where the longest day is 20 hours.[55,56]

Ar. p. 26

They stated that on its course from east to west the line of equality on the earth is cut, between India and Ethiopia, at an island there on the south side of the equator; this island lies across it there and separates north from south. The line that cuts this line from north to south lies halfway between the islands said to be in the Ocean and the farthest inhabited land of China; this is the dome of the earth, known by what we have described, and its place is the place of the intersection. The latitude from the line of equality to the island of Thule is about 60 degrees, a sixth of the circle of the earth. If this sixth, the amount of the latitude, is multiplied by the half, the amount of the longitude, the inhabited land that appears in the north is half a sixth of the earth, that is, one part in twelve.[57,58,59]

They measured the Indian Sea. They said that its length, counted from west to east, from the farthest end of Ethiopia to the farthest end of India, is 8000 miles, and its breadth 2700 miles, of which it extends 1900 miles south of the island where night and day are equal. It has a gulf in the land of Ethiopia that reaches towards al-Barbar and is called the Berber Gulf; its length is 500 miles and the breadth of its channel 100 miles. Another gulf goes out towards the land of Ayla: this is the Sea of al-Qulzum. Its length is 1400 miles; the breadth of its channel, which is called the Green Sea, is 200 miles, and its breadth at its mouth 700 miles. Another gulf goes out towards the land of Persia; it is called the Persian Gulf, and it is the Sea of Basra. Its length is 1400 miles, its breadth at its mouth 500 miles, and the breadth of its channel 150 miles. Between these two gulfs, I mean the gulf of Ayla and the gulf of Persia, lie the lands of the Ḥijāz and Yemen; the distance between these two gulfs is 1500 miles. Another gulf also goes out from it, near its end,

[53] The Arabic writes 78° 28′ and 11° 32′ in abjad numerals; Nallino's Latin prints 78° 23′ in the first place, a misprint. Ptolemy (*Almagest* II 6) puts these two parallels at 69 1/2° and 78 1/3°, because his inclination is larger (Nallino).

[54] The rest of the chapter was edited and translated into French by Reinaud (1848); much of it recurs, nearly word for word, in al-Masʿūdī, Ibn Rusta and Qudāma (Nallino).

[55] The codex has «the empty islands» (*al-khāliyāt*); Nallino reads «the Eternal Islands» (*al-khālidāt*), as Plato of Tivoli and the Arabic geographers have it. They are the Fortunate Islands of the Greek geographers, the Canaries.

[56] 13,500 miles for 180° make 75 miles to the degree, a value that reached the Arabs through Syriac writers from Ptolemy (Nallino).

[57] Ethiopia (*al-Ḥabash*) is here, as in the Greek geographers, the whole south of Africa (Nallino).

[58] The dome of the earth (*qubbat al-arḍ*) goes back to the Indian prime meridian through Ujjain and Laṅkā, misplaced by the Arabs on an island on the equator; it is the «Arin» of the medieval Latin astronomers (Nallino).

[59] The computation is wrong twice over (Schiaparelli, in Nallino's commentary). The area of a zone of the sphere is not the product of its extent in latitude and its extent in longitude. And even on that rule, 60° is a third of the 180° from pole to pole, which would give 1/6, not 1/12. In fact the half of the zone between the equator and latitude 60° is (sin 60°)/4 ≈ 0.2165 of the sphere.

towards the farthest land of India; it is called the Green Gulf, and its length is 1500 miles.[60]

In this whole sea, I mean the Sea of India and China, there are 1370 islands, inhabited and uninhabited. Among them is an island at its far end, near the land of China, called Taprobane, which is Sarandīb; it is 3000 miles round. It faces India on the east. It has great mountains and many rivers, from which come rubies and sapphires. Around it are 59 inhabited islands, with many towns and villages.[61,62]

As for the western Ocean, which is called the Encircling [Sea], only its western and northern part is known, from the farthest land of Ethiopia to Britain; it is a sea on which ships do not sail. The six islands in it, facing the land of Ethiopia, are the inhabited islands, which are also called the Islands of the Blessed. Another island, facing al-Andalus and called Gadira, lies at the strait. This strait goes out from the Ocean; it is 7 miles wide where it goes out, it lies between al-Andalus and Tangier, it is called Sabṭā, and it goes out into the Sea of the Rūm. In the Ocean, on the north side, are also the islands of Britain, which are twelve islands. Beyond them it goes far away from the inhabited land, and no one knows what it is like or what it holds.[63]

Ar. p. 27

The Sea of the Rūm and of Egypt begins at the strait that goes out from the western Ocean near the island called Gadira, facing al-Andalus, and runs east to Tyre and Sidon. Its length is 5000 miles; its breadth is 600 miles in one place, 700 miles in another and 800 miles in another. It has one gulf that goes out towards the north near Rome, 500 miles long, called Adris, and another gulf that goes out towards the land of Narbonne, 200 miles long. In this whole sea there are 162 inhabited islands, five of them large: the island of Corsica, 200 miles round; Sardinia, 300 miles round; Cyprus, 350 miles round; Sicily, 500 miles round; and Crete, 300 miles round.[64]

The Sea of Pontus extends from Lādhiqa to Constantinople the Great; its length is 1060 miles and its breadth 300 miles. The river called Tanais flows into it; it comes from the north, from the lake called Maeotis, which is a large sea although it is called a lake: its length from east to west is 300 miles and its breadth 100 miles. At Constantinople a channel bursts out of the Pontus, flowing like a river, and pours into the Sea of Egypt; at Constantinople it is about 3 miles wide, and Constantinople lies on it.[65]

The Sea of Jurjān, which is the Sea of al-Bāb, is 800 miles long from west to east and 600 miles wide. In it, facing Jurjān, are two islands that were inhabited in the past. These are the inhabited places in the known seas of the earth; God knows best.[66]

The earth has been divided into three parts. The first of them is [bounded] by the Green Sea on the north side, by the channel that goes out from the Pontus into the Greatest Sea, and by what lies between the lake Maeotis and the Pontus. So the boundaries of this region are, on the west and the north, the western sea, which is the Ocean; on the south, the Sea of Egypt and of the

[60] The measures of the seas go back, through a Syriac source also used by Jacob of Edessa, to Ptolemy's map: at 75 miles to the degree they agree with it (Nallino). Al-Barbar is the country of Berbera, on the coast of Somalia. The Sea of al-Qulzum is the Red Sea: al-Qulzum is the Greek Clysma, near Suez, and Ayla is the present al-ʿAqaba. Nallino takes the Green Gulf to be Ptolemy's Prasōdēs Sea, the «leek-green sea» (*Geography* VII 2, 1).

[61] Taprobane and Sarandīb are names of Sri Lanka. «Rubies and sapphires» is literally «red *yāqūt* and the colour of the sky»: *yāqūt* is corundum, red in the ruby and blue in the sapphire.

[62] Jacob of Edessa, Ibn Rusta and al-Kharaqī give 19 islands, the number that Ptolemy lists around Taprobane (*Geography* VII 4, 11–13); Nallino concludes that al-Battānī's source had 19, written in letters (يط), which he read as 59 (نط).

[63] Gadira is Cadiz, and Sabṭā is Ceuta; the strait is the Strait of Gibraltar. The Sea of the Rūm (the Byzantines) is the Mediterranean.

[64] Adris is the Adriatic. Corsica is called by its Greek name, Kyrnos (*Qurnus*).

[65] The Pontus is the Black Sea, the Tanais the Don and the Maeotis the Sea of Azov; the channel is the Bosporus. Lādhiqa, also written Lāziqa, is a city or region of the Caucasus (Nallino). For the length of the Pontus Nallino prefers 1300 miles, the figure of Jacob of Edessa, Ibn Rusta and others.

[66] The Caspian Sea; al-Bāb is Derbent. For its length Nallino prefers 1800 miles, the figure of Ibn Rusta and Jacob of Edessa.

Rūm; and on the east, the Tanais and the lake Maeotis. This land is like an island; they called it Europe.[67]

The second part lies to the south, from the Sea of Egypt to the Sea of Ethiopia. The boundaries of this region are, on the west, the Green Sea; on the north, the Sea of Egypt and of the Rūm; on the east, al-ʿArīsh; and on the south, the Sea of Ethiopia. This part is called Libya.

Ar. p. 28

The third part is all the rest of the inhabited earth, to its farthest end. Its boundaries are, on the west, the Tanais, the river, the channel, al-ʿArīsh and Ayla; on the south, the Sea of Yemen and of India; and on the east, the farthest inhabited land of China towards the east, and China itself. This part is called Greater Asia. These three parts take in the climates, the districts and all the inhabited regions.[68]

What is not known to be either inhabited or desolate is 11 parts out of 12. The part that holds the known inhabited land, from the place of the line of equality, also holds the seas and the deserts. If someone asks whether there are plants, animals and habitation in these eleven parts, the answer can only come from inference by analogy and from opinion. The inhabited earth on our side does not go beyond the limits and boundaries we have mentioned, and no one has brought us word of what lies beyond them. But opinion and conjecture, reasoning by analogy, arrive at what no one of understanding will deny. The sun, the moon and the stars move with us, and from their motion, their nearness and their distance come summer and winter, plants, animals and habitation, and all that everyone knows. If the sun and the stars rise over every place of the rest of the circle of the earth as they do with us, then there can be plants, animals, seas and mountains there as with us; and so it should be. One degree comes to about 65 of the miles mentioned above, which is about two days' journey; God knows best.[69]

As for the longitudes and latitudes of the cities as they are set down in the *Book of the Image of the Earth*: their positions in longitude, which is the distance between west and east, were counted from the inhabited islands in the western Ocean towards the east. This was done according to what they found of the times of eclipses, those of the moon in particular, which occur earlier in some localities than in others. From this they learned that noon in each locality precedes noon in another locality to the west of it by as many times of the equator as lie between [the times of] the eclipse in the two localities. Some of it they took, approximately, from the reports of travellers.[70]

The latitudes of the cities they took from measuring the sun at noon in the localities. They found its distance from the zenith, as we have explained earlier in this book, and so learned the distance of each locality from the line of equality, which is the distance between south and north. Under each city they set down, approximately, its distance from the Eternal Islands in longitude and from the line of equality in latitude. We have entered these according to the description that we found in the well-known *Book of the Image of the Earth*, and we have also listed separately the middles of the known regions and districts, as Ptolemy did; they are 94 regions. There are errors in the longitudes and latitudes in that book; we shall come back to what is needed of this in the later parts of our book.[71]

[67] The Green Sea is here the Atlantic, as often in Arabic authors, and the Greatest Sea the Mediterranean, the «Great Sea» of Orosius (Nallino, after Reinaud).

[68] The river is the channel at Constantinople that flows like a river; the channel is the Sea of the Rūm, which reaches Tyre and Sidon (Nallino).

[69] The 13,500 miles of the half circumference give 75 miles to the degree. Nallino takes 65 for a slip for 75: the Arabic words for sixty and seventy are easily confused when their dots are left out. Plato of Tivoli has 85.

[70] The *Book of the Image of the Earth* (*Kitāb ṣūrat al-arḍ*) is not al-Khwārizmī's book of that title: Nallino shows that its author used both Ptolemy's *Geography* and al-Khwārizmī's book, and suggests that it was the Arabic recension of the *Geography* made by Thābit ibn Qurra. Plato of Tivoli's Latin adds «which is called Geography».

[71] Nallino could not find such a list in Ptolemy: the *Geography* gives no coordinates for the middles of its provinces.

Chapter 7. On finding the winter and summer risings and settings on the horizon circles of the localities, from the excess of the longest day or from the altitude of the pole, whichever of the two is known

Ar. p. 29

He said: Suppose you wish to find the amounts of the arcs of the horizon circle that lie between the equator and the ecliptic at the horizon in any locality; this is the azimuth of the rising and of the setting of any part of the ecliptic you want. Take the given excess of the longest day and find its amount in degrees, by multiplying each hour of it by 15 degrees. Take half the result and add it to 90; the sum is half the arc of the longest day. Then take the whole inclination, which is the declination of the point of the beginning of Cancer, and subtract it from 90. Find the sine of the remainder: this is the sine of the complement of the declination of Cancer. Multiply it by the sine of half the arc of the longest day, divide the product by the radius, and find the arc of the quotient. Subtract this arc from 90; the remainder is the amount [of the arc] on the horizon circle between the rising and the setting of the beginning of Cancer and the equator, to the north of the equator.[72]

Likewise, if you work with half the arc of the shortest day, which is the day of the beginning of Capricorn, the amount comes out the same, and it is known that the beginning of Capricorn rises and sets to the south of the equator. Clearly the azimuth of the beginning of Cancer in the north is equal to the azimuth of the beginning of Capricorn in the south. These are the summer and winter risings and settings: those on the north side are called the summer ones, and those in the south the winter ones.

Ar. p. 30

If you wish to find the azimuth of the rising and setting of points of the ecliptic other than these two, take the declination of any degree you like, subtract it from 90, and find the sine of the remainder. Then find half the arc of the day of that degree, or take it to be whatever you like; find its sine and multiply it by the sine of the complement of the declination of the degree. Divide the product by the radius, find the arc of the quotient, and subtract it from 90; the remainder is the azimuth of the rising and setting of that degree on the horizon circle. If the declination is northern, it lies to the north of the equator; if it is southern, to the south.[73]

If the latitude of the locality is given and you wish to find from it the azimuth of the rising and setting of any degree you like, subtract the latitude of the locality from 90; the remainder is the altitude of the beginning of Aries there. Find its sine. Then take the declination of the degree you want and find its sine; multiply it by the radius and divide the product by the sine of the altitude of the beginning of Aries. Find the arc of the quotient: this arc is the distance of the rising and setting of that degree from the rising and setting of the beginning of Aries, on the horizon circle, towards the side of the declination. This is the azimuth, if God wills.[74]

Chapter 8. On finding the altitude of the pole from the excess of the longest day

Ar. p. 30

He said: Suppose you wish to find the altitude of the north pole of the equator above the horizon, which is the latitude of the locality, from the excess of the longest day over the equinoctial day, or from the shortfall of the shortest day. Take half the excess of the longest day, the day of the first degree of Cancer, in degrees, and add it to 90; the sum is half the arc of the longest day. Or, if you wish, subtract it from 90; the remainder is half the arc of the shortest day. Whichever of the two you work with, it comes to the same. Then subtract the whole inclination from 90 and find the sine of the remainder: this is the sine of the complement of the whole inclination. Multiply

[72] The azimuth of a rising (*samt al-maṭlaʿ*) is measured on the horizon from the point where the equator rises; in modern terms it is the amplitude of the rising. The excess of the longest day is its excess over 12 hours. In modern terms the rule is $\cos A = \cos \delta \cdot \sin H$, where A is the amplitude, δ the declination and H half the arc of the day.

[73] «Half» is supplied by Nallino; the codex and Plato of Tivoli have «the arc of the day».

[74] In modern terms, $\sin A = \sin \delta / \cos \varphi$, where φ is the latitude; $90^\circ - \varphi$ is the altitude of the beginning of Aries when it culminates.

the sine of half the arc of the day by the sine of the complement of the whole inclination, divide the product by the radius, and find the arc of the quotient. The arc you obtain is the distance of the rising of the beginning of Cancer from the north point. Subtract it from 90; the remainder is the distance of the rising of the beginning of Cancer from the equinoctial rising point, as we have explained in the preceding chapter. Then multiply the sine of half the excess of the longest day by the sine of the distance of the rising of the beginning of Cancer from the north point, and divide the product by the sine of the distance of the rising of the beginning of Cancer from the rising of the beginning of Aries. Multiply the quotient by the radius, divide the result by the sine of half the arc of the longest day, and find the arc of the quotient. The arc you obtain is the altitude of the pole where the excess of the longest day is the given amount with which you worked.[75,76]

Chapter 9. On finding the excess of the longest day, and the lesser excesses of the day, from the altitude of the pole

Ar. p. 31

He said: Suppose you wish to find the amount of the excess of the longest day, and of the shortfall of the shortest day, with respect to the equinoctial day, from the altitude of the pole when it is given. Take the latitude of the locality and find its sine. Then subtract the latitude of the locality from 90 and find the sine of the remainder: this is the sine of the complement of the latitude of the locality. Then find the sine of the whole inclination, and the sine of what the whole inclination falls short of 90. Multiply the sine of the latitude of the locality by the sine of the whole inclination, divide the product by the sine of the complement of the whole inclination, multiply the quotient by the radius, divide the result by the sine of the complement of the latitude of the locality, and find the arc of the quotient. The arc you obtain is half the excess of the longest day, and half the shortfall of the shortest day is the same. Double it: the result is the whole excess of the longest day and the whole shortfall of the shortest day. Every 15 degrees of it make one equal hour. Add the hours you obtain to the 12 hours that are the length of the equinoctial day; the sum is the hours of the longest day. Subtract the same excess from 12; the remainder is the hours of the shortest day.[77]

If you wish to know the excess of the day at degrees of the ecliptic other than these two, take the declination of any degree of the signs you like and work with it in place of the whole inclination. What you obtain at the end of the work, done in the same way, is the difference of the day at that degree: if the declination of the degree is northern, it is an excess of the day; if it is southern, it is a shortfall at that degree.

[75] The equinoctial day (*al-nahār al-muʿtadil*) is the day of 12 hours. In modern terms the rule is $\sin \varphi = \sin e \cdot \cos A / (\sin A \cdot \sin H)$, where φ is the latitude, e half the excess of the longest day, $H = 90° + e$ half the arc of the longest day, and A the amplitude of the rising of the beginning of Cancer. The rule is exact, since $\sin e = \tan \varphi \cdot \tan \varepsilon$, $\sin A = \sin \varepsilon / \cos \varphi$ and $\cos A = \cos \varepsilon \cdot \sin H$.

[76] Here the codex writes 90 in Indian figures, «٩٠» (Nallino: «so the codex itself»); elsewhere the text writes numbers in words or in abjad letters.

[77] In modern terms, $\sin e = \tan \varphi \cdot \tan \varepsilon$, where e is half the excess of the longest day, φ the latitude and ε the inclination.